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To cite this article: Nicolas W. Medrano, Cynthia Lizette Villarreal, N. Clay Mann, Michelle A. Price, Kurt B. Nolte, Ellen J. MacKenzie, Pam Bixby, Brian J. Eastridge & for the MIMIC Study Group (2023) Activation and On-Scene Intervals for Severe Trauma EMS Interventions: An Analysis of the NEMSIS Database, *Prehospital Emergency Care*, 27:1, 46-53, DOI: [10.1080/10903127.2022.2053615](https://doi.org/10.1080/10903127.2022.2053615)

To link to this article: <https://doi.org/10.1080/10903127.2022.2053615>



Published online: 15 Apr 2022.



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Activation and On-Scene Intervals for Severe Trauma EMS Interventions: An Analysis of the NEMSIS Database

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ABSTRACT

Objective: Time to care is a determinant of trauma patient outcomes, and timely delivery of trauma care to severely injured patients is critical in reducing mortality. Numerous studies have analyzed access to care using prehospital intervals from a Carr et al. meta-analysis of studies from 1975 to 2005. Carr et al.'s research sought to determine national mean activation and on-scene intervals for trauma patients using contemporary emergency medical services (EMS) records. Since the Carr et al. meta-analysis was published, the National Highway Traffic Safety Administration (NHTSA) created and refined the National Emergency Medical Services Information System (NEMSIS) database. We sought to perform a modern analysis of prehospital intervals to establish current standards and temporal patterns.

Methods: We utilized NEMSIS to analyze EMS data of trauma patients from 2016 to 2019. The dataset comprises more than 94 million EMS records, which we filtered to select for severe trauma and stratified by type of transport and rurality to calculate mean activation and on-scene intervals. Furthermore, we explored the impact of basic life support (BLS) and advanced life support (ALS) of ground units on activation and on-scene time intervals.

Results: Mean activation and on-scene intervals for ground transport were statistically different when stratified by rurality. Urban, suburban, and rural ground activation intervals were 2.60 ± 3.94 , 2.88 ± 3.89 , and 3.33 ± 4.58 minutes, respectively. On-scene intervals were 15.50 ± 10.46 , 17.56 ± 11.27 , and 18.07 ± 16.13 minutes, respectively. Mean helicopter transport activation time was 13.75 ± 7.44 minutes and on-scene time was 19.42 ± 16.09 minutes. This analysis provides an empirically defined mean for activation and on-scene times for trauma patients based on transport type and rurality. Results from this analysis proved to be significantly longer than the previous analysis, except for helicopter transport on-scene time. Shorter mean intervals were seen in ALS compared to BLS for activation intervals, however ALS on-scene intervals were marginally longer than BLS.

Conclusions: With the increasing sophistication of geospatial technologies employed to analyze access to care, these intervals are the most accurate and up-to-date and should be included in access to care models.

ARTICLE HISTORY

Received 31 July 2021
Revised 10 March 2022
Accepted 11 March 2022

Introduction

Injury remains the leading cause of death in individuals up to the age of 44 and the leading cause of morbidity and mortality among children in the U.S. (1). Of these trauma deaths, between 200,000 and 300,000 over 10 years would be potentially preventable if the patients received rapid and high-quality trauma care (2, 3). Furthermore, a 2016 report from the National Academies of Science, Engineering and Medicine (NASEM), entitled “A National Trauma Care System: Integrating Military and Civilian Trauma Systems to Achieve Zero Preventable Deaths After Injury,” estimated that approximately 30,000 of the 147,790 people who died as a result of trauma in 2014 had potentially survivable injuries (4). During the last several decades, advances in care in trauma centers and across trauma systems have substantially

reduced death and disability associated with injury (5, 6). However, there remains a substantial opportunity to further reduce the number of deaths in the prehospital setting.

The first hour after traumatic injury is historically known as the “golden hour” and is the period of care following an injury in which rapid assessment and resuscitation are fundamental to survival (7). Though there is no true golden hour, the time lapse between injury and care is a critical element for patients with severe injury (8–12), with longer prehospital transport times likely contributing to higher mortality rates in more severely injured patients. This effect is more prominent among rural trauma patients than similarly injured urban patients (13–17). To substantiate this assertion, it has been demonstrated that states with poor trauma care access have more prehospital deaths,

Table 1. Data elements used for trauma case selection.

Data element	V2 code	V3 code	Code description
Type of service	30	2205001	911 response
Response mode to scene	390	2224015	Lights and sirens
Complaint reported by dispatch	410	2301005	Animal bite
	415	2301007	Assault
	N/A	2301009	Automated crash notification
	430	2301015	Burns
	470	2301029	Electrocution
	475	2301031	Eye problem
	480	2301033	Fall victim
	495	2301045	Heat/cold exposure
	500	4301043	Hemorrhage/laceration
	505	2301047	Industrial accident
	530	2301063	Stab/gunshot wound
	540	2301069	Traffic accident
	545	2301073	Traumatic injury
Possible injury	1	9922005	Yes
Transport mode from scene	4965	4218011	Lights and sirens
Type of destination	7280	4221003	Hospital
Unit transport and equipment	7340	4221015	Other EMS responder
	1030	2207011	Air transport-helicopter
		2207003	Ground transport
	990	2215001	BLS – first responder
	995	2215003	BLS – basic/EMT
		2215005	BLS – AEMT
		2215007	BLS – intermediate
		2215023	BLS – community paramedicine
	1000	2215009	
	1005	2215011	ALS – AEMT
	1010	2215013	ALS – intermediate
		2215015	ALS – paramedic
		2215017	ALS – community paramedicine
		2215019	ALS – nurse
			ALS – physician

contributing to higher overall injury mortality (9). Because of the “access effect” on mortality and morbidity rates, having an accurate understanding of prehospital intervals is imperative to identifying geographic “trauma care deserts,” where there is decreased access to immediate care.

Numerous researchers have created models to analyze access to trauma care in recent years, breaking the total prehospital time into four time intervals: activation, response, on-scene, and transport. The activation interval is defined as the time from emergency call to ambulance dispatch. The response interval is the time from ambulance dispatch to arrival at the scene. The on-scene interval is the time from ambulance arrival at the scene to the time the ambulance departs from the scene. Finally, the transport interval is the time from ambulance departure from the scene to arrival at the hospital. Many of the models created to estimate access to care have used mean activation and on-scene intervals calculated from a 2005 meta-analysis conducted by Carr et al. (18). Some models combine mean activation and on-scene times from this meta-analysis with mean driving speeds based on rurality (19–22), while more recent studies have begun utilizing geographic information systems (GIS) and road networks built with actual traffic data (23–26). Results from the 2005 meta-analysis (18) of 49 observational studies have been immensely valuable; however, we believe mean activation and on-scene intervals derived from more current datasets will be more applicable to the current era of trauma care and systems and would better support future access to care research. This study aimed to recalculate and update activation and on-scene prehospital intervals using datasets from a single data source, the National

Emergency Medical Services Information System (NEMSIS). In doing so, our goal was to produce empirically derived prehospital means from more contemporary data to support future access to care models and provide a blueprint for future utilization of the NEMSIS dataset for time analyses.

Methods

Data Collection

National Emergency Medical Services Information System records were obtained for 2016–2019, accounting for approximately 94.5 million EMS events over four years. The NEMSIS database is a product of the National Highway Traffic Safety Administration’s (NHTSA) Office of EMS, which uses a universal standard for patient care information resulting from an emergency 911 call for assistance. The NEMSIS is a collaborative system created to improve patient care through standardization, aggregation, and utilization of point of care EMS data at a local, state, and national level. The data was submitted by more than 11,000 participating EMS agencies in 49 states. In 2017, the NEMSIS Version 2 data dictionary standard officially closed, and states migrated to Version 3. Therefore, this analysis comprises data from Version 2 and Version 3 data dictionaries (27).

Case Selection

Each year’s dataset was imported into SAS 9.4 for analysis, where data elements were used to select EMS responses for

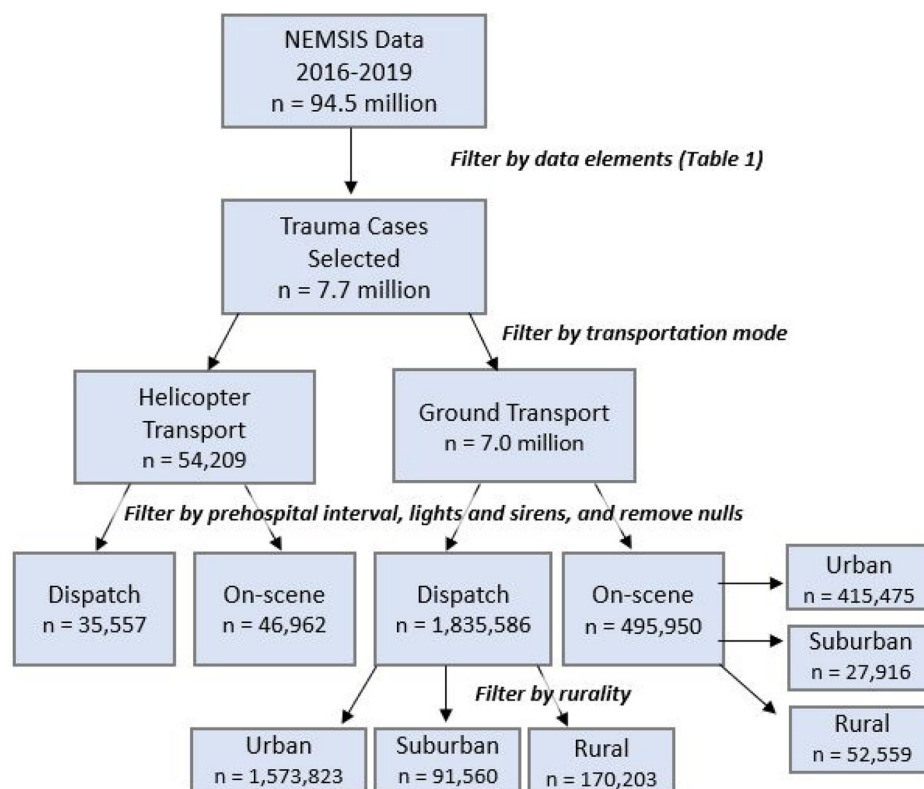


Figure 1. Filter diagram of NEMSIS data.

severe traumatic injury. We defined trauma cases through a combination of data elements: type of service, response mode to scene, complaint reported by dispatch, possible injury, transport mode from scene, and type of destination (Table 1). Lights and sirens being utilized for response mode to scene and transport mode from scene was used to identify time-sensitive severe trauma cases where EMS responders facilitated patient transport in an urgent manner.

After the selection of severe trauma cases, records were filtered by transportation type, level of care basic life support (BLS) and advanced life support (ALS), and rurality. Due to their lower frequency, helicopter transport cases were not stratified by rurality. Rurality in NEMSIS is categorized by USDA Urban Influence Codes that form a classification scheme distinguishing metropolitan counties by population size of the metro area, and nonmetropolitan counties by the size of the largest city or town and proximity to metro areas (28). Ground transport cases were stratified by rurality into urban, suburban, and rural cohorts. This is an important qualification, as EMS infrastructure, resources, and operations vary significantly among these environments. Events coded as frontier/wilderness were also classified as rural. Finally, the NEMSIS data element *EMSSceneTime* was used for mean on-scene interval; and the elapsed time intervals *EMSDispatchCenterTime* and *EMSchuteTime* were summed for the mean activation interval. *EMSDispatchCenterTime* represents the time difference between when the 911 call is received by dispatch and when the ambulance unit is notified. *EMSchuteTime* represents the time difference between when the ambulance unit is notified and when it is en route. For activation intervals, lights and sirens were required for

the response interval to meet inclusion criteria. For the on-scene interval, lights and sirens were required for the transport interval to meet inclusion criteria (Figure 1).

Data Analysis

Case removal was required for records with null values for any time element, times less than or equal to zero, and outliers. Derived elapsed time elements were excluded from the analysis if outside the standardized range set by NEMSIS (29). This includes dispatch times greater than, or equal to, one hour and on scene times greater than, or equal to, one day. Median, mean and standard deviation was calculated for both activation intervals, and on-scene intervals, and Student's *t*-Tests were performed for statistical comparisons between transportation type and rurality. Finally, an Independent Samples *t*-Test was used to compare this NEMSIS analysis with Carr et al.'s meta-analysis results.

Results

In total, approximately 1.9 million records were identified for activation interval analysis, represented by 35,557 air transport and 1,835,586 ground transport activations. Urban EMS activations accounted for most ground transport findings, with 1,573,823 events. For the on-scene interval, 542,912 EMS activations were identified, represented by 46,962 helicopter transport and 495,950 ground transport events. Urban EMS on-scene events accounted for most ground transport findings, with 415,475 events. The EMS activations provided from the NEMSIS analysis were more

Table 2. Number of activations for prehospital intervals of helicopter and ground transport of trauma patients.

	Helicopter transport	Urban ground transport	Suburban ground transport	Rural ground transport
Activation interval				
NEMSIS	35,557	1,573,823	91,560	170,203
Carr et al	2,281	105,145	105,145	6,846
On-scene interval				
NEMSIS	46,962	415,475	27,916	52,559
Carr et al	4,047	139,866	127,850	30,047

robust than Carr et al., except for suburban ground data (Table 2).

Helicopter transport mean activation and on-scene time intervals were significantly longer than those for ground transport, regardless of rurality ($p < 0.01$) (Table 3). Statistically significant differences were noted between all mean ground intervals, with urban total activation and on-scene intervals being significantly shorter than suburban and rural ($p < 0.01$). Furthermore, while *EMSDispatchCenterTime* was quite similar between all ground units, *EMSchuteTime* varied, with rural being more than two times longer than urban.

When comparing NEMSIS data to the meta-analysis data, there were mixed results. Mean activation intervals for all NEMSIS data were significantly longer than those from Carr et al. ($p < 0.01$) (Table 3). The difference in the helicopter transport mean activation interval was quite stark. The NEMSIS interval was more than three times longer than the interval calculated from Carr et al. Alternatively, the mean helicopter interval for on-scene time was significantly less for the NEMSIS data ($p < 0.01$). All other NEMSIS mean on-scene intervals were significantly longer than those provided by Carr et al. ($p < 0.01$).

An extensive analysis evaluating the impact of advanced life support (ALS) and basic life support (BLS) of ground units showed mixed results. Shorter mean times were seen in ALS compared to BLS for all activation intervals across all ruralities ($p < 0.01$). The largest difference between ALS and BLS for the total activation interval was seen in the rural environment (3.10 ± 4.31 and 4.84 ± 5.86 respectively). Conversely, overall mean on-scene intervals were quite similar, with marginally longer intervals for ALS ($p < 0.01$). Intervals between ALS and BLS were quite similar in urban environments (15.54 ± 10.73 and 15.38 ± 9.47 minutes respectively) but varied substantially for suburban (18.02 ± 11.60 and 16.03 ± 9.93 minutes) and rural (18.67 ± 15.18 and 16.31 ± 18.57 minutes) responses (Table 4).

Discussion

Our analysis of EMS elapsed service times calculated activation and on-scene prehospital intervals using datasets from a single modern and uniform data source. These empirically derived prehospital intervals provide valuable data for future access to care models and serve as a blueprint for future utilization of the NEMSIS dataset. With longer prehospital times contributing to higher mortality rates in severely injured patients, making sure patients have timely access to trauma care is an essential component of reaching the goal of “Zero Preventable Deaths After Injury” set by the NASEM report (4). Therefore, accurately estimating trauma

care accessibility is of utmost importance when identifying trauma care deserts, where timely access to care is absent.

Prehospital Intervals

Surprisingly, except for helicopter transport on-scene times, NEMSIS time intervals were significantly longer than results from the meta-analysis completed by Carr et al. We believe this may be due to a few reasons. First, as stated in the analysis by Carr et al., “... strict definitions of prehospital time intervals and conformity to existing definitions could not be controlled as inconsistencies exist with the literature reviewed for this analysis.” This limitation should be minimized with the current analysis due to the standardization present within the NEMSIS database. Also, all cases with a time of zero were removed from this analysis, as these are not possible or useful when determining mean time intervals. If an EMS provider reported the same time for arrival and departure from a scene, we considered this a data reporting error or an instance when EMS did not provide aid. It is possible that some articles in the meta-analysis included times of zero and, therefore, lowered the mean time intervals. Second, the significantly longer times calculated from the NEMSIS dataset are due to a difference in the activation interval definition. According to Carr et al., the activation interval is defined as “the time from call was received to time of alarm.” This assumes the ambulance is en route to the scene instantaneously at the time of notification. The two-component time definition of the activation interval is more accurate and realistic because it includes the time from the 911 call to unit notification and the time between when the unit is notified and when the unit is en route.

While all activation and on-scene times from our analysis are statistically different from one another, it was quite surprising to see such similarities, especially regarding all ground unit *EMSDispatchCenterTime* and on-scene time between suburban and rural communities. Furthermore, analysis of the two activation components suggests the activation interval variability among ruralities is mainly due to *EMSchuteTime*, the time from EMS notification until the unit is en route. We suspect this difference is due to the volunteer-based systems more commonly found in rural communities. Finally, if total activation and on-scene intervals are summed, more stark differences between the ruralities begin to emerge.

The analysis of the NEMSIS dataset showed considerable differences in activation intervals, but on-scene intervals were comparable. There was a difference of 1.47 minutes between BLS and ALS for total activation time, with ALS

Table 3. Mean and standard deviations for activation and on-scene intervals (mins) of helicopter and ground units for trauma responses.

	Helicopter transport	Urban ground transport	Suburban ground transport	Rural ground transport
Activation Interval				
NEMSIS				
<i>EMSDispatchCenterTime</i>	5.19 ± 6.79	1.59 ± 3.65	1.55 ± 3.31	1.36 ± 3.40
<i>EMSchuteTime</i>	8.57 ± 4.90	1.01 ± 1.56	1.34 ± 2.14	1.97 ± 3.00
Total activation interval	13.75 ± 7.44	2.60 ± 3.94	2.88 ± 3.89	3.33 ± 4.58
Median	12.00	1.68	2.00	2.00
Carr et al.	3.53 ± 3.81	1.40 ± 1.41	1.40 ± 1.41	2.89 ± 1.64
On-Scene Interval				
NEMSIS				
Median	16.00	14.00	16.00	15.92
Mean & SD	19.42 ± 16.09	15.50 ± 10.46	17.56 ± 11.27	18.07 ± 16.13
Carr et al.	21.60 ± 18.90	13.50 ± 3.71	13.45 ± 21.80	15.06 ± 16.80

Table 4. Mean and standard deviations for activation and on-scene intervals for BLS and ALS ground units for trauma responses.

	Urban BLS	Urban ALS	Suburban BLS	Suburban ALS	Rural BLS	Rural ALS	All BLS	All ALS
Activation Interval								
NEMSIS								
<i>EMSDispatchCenterTime</i>	2.48 ± 5.32	1.49 ± 3.40	1.96 ± 3.68	1.50 ± 3.27	1.61 ± 3.92	1.31 ± 3.32	2.35 ± 5.11	1.47 ± 3.39
<i>EMSchuteTime</i>	1.37 ± 2.15	0.97 ± 1.48	2.35 ± 3.56	1.22 ± 1.87	3.22 ± 4.26	1.78 ± 2.70	1.64 ± 2.65	1.06 ± 1.66
Total activation interval	3.86 ± 5.60	2.46 ± 3.69	4.30 ± 4.80	2.72 ± 3.73	4.84 ± 5.86	3.10 ± 4.31	4.00 ± 5.60	2.53 ± 3.75
Median	2.18	1.62	3.00	2.00	3.00	2.00	2.32	1.68
<i>n</i>	157,853	1,415,970	9,540	82,020	22,746	147,457	190,139	1,645,447
On-Scene Interval								
Median	13.92	14.00	14.00	16.00	14.00	16.00	14.00	14.25
Mean & SD	15.38 ± 9.47	15.54 ± 10.73	16.03 ± 9.93	18.02 ± 11.60	16.31 ± 18.57	18.67 ± 15.18	15.52 ± 10.93	16.00 ± 11.38
<i>n</i>	96,402	319,073	6,394	21,522	13,296	39,263	116,092	379,858

being 58% faster than BLS at getting en route to the scene. In urban environments, this discrepancy was driven by *EMSDispatchCenterTime*, while in suburban and rural environments it was caused more by *EMSchuteTime*. It is possible that the more expeditious ALS response were due to differences in injury severity, with ALS being utilized more frequently for severe injuries, and therefore urgency is prioritized. While total on-scene times were very similar, on-scene time for ALS was longer than BLS for all ruralities. These differences are very minor in urban environments but become substantial in suburban and rural locations. We believe this is due to the increased capabilities and invasive methods used by ALS.

Limitations

There are limitations associated with the NEMSIS dataset that must be acknowledged. First, the dataset is a large convenience sample, consisting solely of data submitted by participating EMS agencies and may not represent the entire U.S. population. It likely includes a disproportionate number of EMS agencies with the resources and leadership necessary to participate in the program. This may explain the large number of urban results compared to suburban and rural environments. While under the Census Bureau designation, roughly 23 percent of the population is considered rural, only approximately 10 percent of the sample was represented by rural EMS providers. Second, while the NEMSIS database contains data from a wide array of states, the authors cannot say for certain how many states are represented once case selection was completed. The NEMSIS public-release research does not disclose the state in which the record was generated. However, results include activations from all nine U.S. Census Divisions, representing a

diverse geographic sample. Third, NEMSIS contains records from two-tier EMS response systems consisting of a first responder crew and transporting crew. In these two-tier systems, the transporting crew is often not required to respond with the same urgency as the first responding crew. There is no definitive way to identify records from these systems, and therefore both are potentially included, resulting in longer dispatch intervals. Finally, due to differences in inclusion/exclusion criteria, potential selection bias could have occurred and affected case selection during the filtering process (29).

Conclusion

As GIS technology is becoming more widely utilized in this field, accurate activation and on-scene intervals are essential for geospatial models analyzing access to trauma care. The values derived from this analysis represent a more accurate and up-to-date representation of prehospital intervals from EMS systems currently in place. This analysis more effectively defines the EMS activation and on-scene intervals by utilizing the national standardized data protocol from EMS agencies across the nation. In addition to establishing the modern baseline metric for these critical prehospital care intervals, these data will be important to identifying and quantifying areas for focused process improvement, enhancing access to care, and assessing the impact of any such interventions by comparing pre and post-intervention interval times.

Disclaimer

The U. S. Army Medical Research Acquisition Activity, 820 Chandler Street, Fort Detrick MD 21702-5014, is the awarding and administering acquisition office. This work was supported by the Office of the

Assistant Secretary of Defense for Health Affairs, through the Defense Medical Research and Development Program under Award No. W81XWH-17-2-0010. Opinions, interpretations, conclusions, and recommendations are those of the author and are not necessarily endorsed by the Department of Defense.

Disclosure Statement

No interest to declare.

Data Availability

The data that support the findings of this study are openly available in the Harvard Dataverse Repository.

Funding

This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs, through the Defense Medical Research and Development Program under Award No W81XWH-17-2-0010.

References

- Centers for Disease Control and Prevention. CDC national health report: leading cause of morbidity and mortality and associated behavioral risk and protective factors - United States, 2005–2013. 2014.
- Hashmi ZG, Haut ER, Efron DT, Salim A, Cornwell EE, III, Haider AH. A target to achieve zero preventable trauma deaths through quality improvement. *JAMA Surg.* 2018;153(7):686–9. doi:10.1001/jamasurg.2018.0159.
- Kwon A, Garbett N, Kloecker G. Pooled preventable death rates in trauma patients: meta analysis and systematic review since 1990. *Eur J Trauma Emerg Surg.* 2014;40(3):279–85. doi:10.1007/s00068-013-0364-5.
- National Academies of Sciences, Engineering, and Medicine. A national trauma care system: integrating military and civilian trauma systems to achieve zero preventable deaths after injury. Washington, DC: National Academies Press; 2016.
- MacKenzie EJ, Rivara FP, Jurkovich GJ, Nathens AB, Frey KP, Egleston BL, Salkever DS, Scharfstein DO. A national evaluation of the effect of trauma-center care on mortality. *N Engl J Med.* 2006;354(4):366–78. doi:10.1056/NEJMsa052049.
- MacKenzie EJ, Weir S, Rivara FP, Jurkovich GJ, Nathens AB, Wang W, Scharfstein DO, Salkever DS. The value of trauma center care. *J Trauma.* 2010;69(1):1–10. doi:10.1097/TA.0b013e3181e03a21.
- Nasr A, Saavedra Tomasich F, Collaço I, Abreu P, Namias N, Marttos A. The trauma golden hour: a practical guide. New York (NY): Springer International Publishing; 2019.
- Sampalis JS, Lavoie A, Williams JL, Mulder DS, Kalina M. Impact of on-site care, prehospital time, and level of in-hospital care on survival in severely injured patients. *J Trauma.* 1993;34(2):252–61. doi:10.1097/00005373-199302000-00014.
- Hashmi ZG, Jarman MP, Uribe-Leitz T, Goralnick E, Newgard CD, Salim A, Cornwell E, III, Haider AH. Access delayed is access denied: relationship between access to trauma center care and pre-hospital death. *J Am Coll Surg.* 2019;228(1):9–20. doi:10.1016/j.jamcollsurg.2018.09.015.
- Feero S, Hedges JR, Simmons E, Irwin L. Does out-of-hospital EMS time affect trauma survival? *Am J Emerg Med.* 1995;13(2):133–5. doi:10.1016/0735-6757(95)90078-0.
- Gervin A, Fischer R. The importance of prompt transport in salvage of patients with penetrating heart wounds. *J Emerg Med.* 1983;22:443–8.
- Alarhayem AQ, Myers JG, Dent D, Liao L, Muir M, Mueller D, Nicholson S, Cestero R, Johnson MC, Stewart R, et al. Time is the enemy: mortality in trauma patients with hemorrhage from torso injury occurs long before the “golden hour”. *Am J Surg.* 2016;212(6):1101–5. doi:10.1016/j.amjsurg.2016.08.018.
- Gonzalez RP, Cummings G, Mulekar M, Rodning CB. Increased mortality in rural vehicular trauma: identifying contributing factors through data linkage. *J Trauma Acute Care Surg.* 2006;61:404–9.
- Grossman DC, Hart LG, Rivara FP, Maier RV, Rosenblatt R. From roadside to bedside: the regionalization of trauma care in a remote rural county. *J Trauma.* 1995;38(1):14–21. doi:10.1097/00005373-199501000-00006.
- Esposito TJ, Maier RV, Rivara FP, Pilcher S, Griffith J, Lazear S, Hogan S. The impact of variation in trauma care times: urban versus rural. *Prehosp Disaster Med.* 1995;10(3):161. doi:10.1017/s1049023x00041947.
- Crandall M, Sharp D, Unger E, Straus D, Brasel K, Hsia R, Esposito T. Trauma deserts: distance from a trauma center, transport times, and mortality from gunshot wounds in Chicago. *Am J Public Health.* 2013;103(6):1103–9. doi:10.2105/AJPH.2013.301223.
- Byrne JP, Mann NC, Dai M, Mason SA, Karanickolas P, Rizoli S, Nathens AB. Association between emergency medical service response time and motor vehicle crash mortality in the United States. *JAMA Surg.* 2019;154(4):286–93. doi:10.1001/jamasurg.2018.5097.
- Carr BG, Caplan JM, Pryor JP, Branas CC. A meta-analysis of prehospital care times for trauma. *Prehosp Emerg Care.* 2006;10(2):198–206. doi:10.1080/10903120500541324.
- Branas C, MacKenzie E, Williams J, Schwab W, Teter H, Flanagan M, Blatt A, ReVelle C. Access to trauma centers in the United States. *JAMA.* 2005;293(21):2626–33. doi:10.1001/jama.293.21.2626.
- Nance ML, Carr BG, Branas CC. Access to pediatric trauma care in the United States. *Arch Pediatr Adolesc Med.* 2009;163(6):512–8. doi:10.1001/archpediatrics.2009.65.
- Carr BG, Branas CC, Metlay JP, Sullivan AF, Camargo CA. Access to emergency care in the United States. *Ann Emerg Med.* 2009;54(2):261–9. doi:10.1016/j.annemergmed.2008.11.016.
- Albright KC, Branas CC, Meyer BC, Matherne-Meyer DE, Zivin JA, Lyden PD, Carr BG. ACCESS: acute cerebrovascular care in emergency stroke systems. *Arch Neurol.* 2010;67(10):1210–8. doi:10.1001/archneurol.2010.250.
- Medrano NW, Villarreal CL, Price MA, MacKenzie E, Nolte KB, Phillips MJ, Stewart RM, Eastridge BJ. Multi-Institutional multidisciplinary injury mortality investigation in the civilian pre-hospital environment (MIMIC): a methodology for reliably measuring pre-hospital time and distance to definitive care. *Trauma Surg Acute Care Open.* 2019;4(1):e000309. doi:10.1136/tsaco-2019-000309.
- Klein MB, Kramer CB, Nelson J, Rivara FP, Gibran NS, Concannon T. Geographic access to burn center hospitals. *JAMA.* 2009;302(16):1774–81. doi:10.1001/jama.2009.1548.
- Nallamothu BK, Bates ER, Wang Y, Bradley EH, Krumholz HM. Driving times and distances to hospitals with percutaneous coronary intervention in the United States: implications for prehospital triage of patients with ST-elevation myocardial infarction. *Circulation.* 2006;113(9):1189–95. doi:10.1161/CIRCULATIONAHA.105.596346.
- Carr BG, Bowman AJ, Wolff CS, Mullen MT, Holena DN, Branas CC, Wiebe DJ. Disparities in access to trauma care in the United States: a population-based analysis. *Injury.* 2017;48(2):332–8. doi:10.1016/j.injury.2017.01.008.
- NEMSIS Version 2 to 3.4.0 Mapping/Translation Tools: National EMS Information System Technical Assistance Center; 2021 [accessed 2021 Jul 30]. <https://nemsis.org/technical-resources/version-3/v3-element-mappingtranslation-documents/>.
- Urban influence codes. In U.S. Department of Agriculture, editor. Washington (DC): Economic Research Service; 2013.
- NEMSIS data user manual: National EMS Information System Technical Assistance Center; 2021. [accessed 2021 Jul 30]. <https://nemsis.org/using-ems-data/request-research-data/>.

Appendix 1

Albrecht, RM	University of Oklahoma Health Sciences Center
Bailey, JA	Uniformed Services University of the Health Sciences
Benjamin, ER	Emory University / Grady Memorial Hospital
Bernard, AC	University of Kentucky
Blackwell, TH	University of South Carolina School of Medicine Greenville
Braithwaite, SA	Washington University in St Louis
Brasel, KJ	Oregon Health & Science University
Brice, JH	University of North Carolina
Bulger, EM	University of Washington
Burlew, CC	Denver Health Medical Center
Butler, FK	DSO Medical Associates
Callaway, DW	Atrium Health
Cannon, JW	University of Pennsylvania
Champion, HR	SimQuest
Chang, M	University of South Alabama
Coimbra, RS	Riverside University Health System & Loma Linda University
Davis, GG	University of Alabama at Birmingham, Jefferson County Medical Examiner Office
Demarest, GB	University of New Mexico School of Medicine
Dorlac, WC	University of Colorado
Drake, SA	Texas A&M University
Eastman, AL	United States Department of Homeland Security
Eastridge, BJ	University of Texas Health Science Center at San Antonio
Elster, EA	Uniformed Services University of the Health Sciences
Epley, E	Southwest Texas Regional Advisory Council
Espósito, TJ	University of Illinois College of Medicine
Ficke, JR	Johns Hopkins University
Fisher, AD	University of New Mexico School of Medicine
Fowler, DR	Maryland Office of the Chief Medical Examiner
Gaines, BA	University of Pittsburgh, Children's Hospital of Pittsburgh
Gallagher, JM	Wichita/Sedgwick County EMS System
Gary, JL	Keck School of Medicine of the University of Southern California
Gestring, ML	University of Rochester
Gill, JR	Connecticut Office of the Chief Medical Examiner
Goodloe, JM	University of Oklahoma School of Community Medicine
Gurney, JM	Joint Trauma System / Institute of Surgical Research
Harrell, AJ	University of New Mexico School of Medicine
Henry, SM	University of Maryland
Holcomb, JB	University of Alabama at Birmingham
Hunt, JL	University of New Mexico
Jenkins, DH	University of Texas Health Science Center at San Antonio
Johannigman, JA	Brooke Army Medical Center
Kerby, JD	University of Alabama at Birmingham
Kharod, CU	Joint Trauma System, Defense Health Agency
Kotwal, RS	Joint Trauma System, Defense Health Agency
Kozar, RA	Shock Trauma, University of Maryland School of Medicine
Kuhls, DA	University of Nevada Las Vegas
Lathrop, SL	University of New Mexico
Latimer, AJ	University of Washington Department of Emergency Medicine
Levy, M	University of Alaska Anchorage
Mabry, RL	Special Operations Command - Fort Bragg
MacKenzie, EJ	Johns Hopkins University Bloomberg School of Public Health
Martin, MJ	Scripps Mercy Hospital
Maxson, RT	Arkansas Children's Hospital
Mazuchowski, EL	Armed Forces Medical Examiner System
Medrano, NW	Coalition for National Trauma Research
Minei, JP	UT Southwestern Medical Center
Mitchell, RA	Washington DC Office of the Chief Medical Examiner
Moore, EE	Ernest E Moore Shock Trauma Center at Denver Health
Moore, LE	Uniformed Services University of the Health Sciences, University of Virginia
Nashelsky, MB	University of Iowa Hospitals and Clinics
Nathens, AB	Sunnybrook Health Sciences Center & University of Toronto
Nolte, KB	University of New Mexico School of Medicine
O'Keefe, GE	University of Washington
Phillips, MJ	Coalition for National Trauma Research
Price, MA	Coalition for National Trauma Research
Robinson, JL	Spectrum Retirement Communities, LLC / International Association of EMS Chiefs
Sagraves, SG	Baylor Scott & White Health
Scalea, TM	University of Maryland
Schenarts, PJ	Creighton University & Des Moines University
Schreiber, MA	Oregon Health & Science University
Shackelford, SA	Joint Trauma System
Sperry, JL	University of Pittsburgh

(continued)

Stassen, NA	University of Rochester
Staudenmayer, KL	Stanford University
Stewart, RM	University of Texas Health Science Center at San Antonio
Stuke, LE	Louisiana State University
Valadka, AB	Virginia Commonwealth University
Villarreal, CL	Coalition for National Trauma Research
Winchell, RJ	Weill Cornell Medicine
Zonies, D	Oregon Health & Science University
Yelon, JA	EMF Great Lakes
